

DAT 490 Capstone: Data Cleaning and Spatial Join

Contents

1	LAPD Data	2
1.1	Load Raw Data	2
1.2	Select Columns	2
1.3	Clean Time Data	2
1.4	Clean Coordinates & Flag Invalid Locations	2
1.5	Remove Duplicates	2
1.6	Missing Values Check	2
1.7	Derive Variables	3
1.8	Export Clean LAPD Data	3
2	ACS Census Data	4
2.1	Load Raw Data	4
2.2	Flag Unreliable Estimates	4
2.3	Reshape Data	5
2.4	Remove Zero Population Tracts	5
2.5	Derive Variables	5
2.6	Check Missing Values	5
2.7	Impute Missing Values	6
2.8	Drop Raw Input Variables	7
2.9	Export Clean ACS Data	7
3	Join and Aggregate Data	8
3.1	Convert LAPD to Spatial Points	8
3.2	Spatial Join	8
3.3	Build RQ1 Incident-Level Dataset	8
3.4	Aggregate to Tract Level for RQ2	9
3.5	Compute Crime Rates	10
3.6	Join ACS Variables	10
3.7	Filter to City of Los Angeles	10
3.8	Tract Flags	11
3.9	Check Join Results	11
3.10	Zero-Incident Tract Details	12
3.11	Remove Zero-Incident and Small Population Tracts	13
4	Variable Selection	14
4.1	Crime Variables	14
4.2	ACS Variables	14
4.3	Export Dataset	16
5	Final Dataset Overview	17
5.1	Tract Flags	17
5.2	Tract-Level Crime Variables (<code>tract_data</code>)	17
5.3	RQ1 Incident-Level Variables (<code>incident_data</code>)	18
5.4	Tract-Level ACS Variables (<code>tract_data</code>)	19

1 LAPD Data

1.1 Load Raw Data

```
## # A tibble: 6 x 28
##   dr_no date_rptd date_occ time_occ area area_name rpt_dist_no part_1_2 crm_cd
##   <dbl> <chr>    <chr>    <chr>    <chr> <chr>    <chr>          <dbl> <dbl>
## 1 2.12e8 04/11/20~ 11/07/2~ 0845     15 N Hollyw~ 1502          2    354
## 2 2.02e8 10/21/20~ 10/18/2~ 1845     15 N Hollyw~ 1521          1    230
## 3 2.41e8 12/10/20~ 10/30/2~ 1240     09 Van Nuys 0933          2    354
## 4 2.11e8 12/24/20~ 12/24/2~ 1310     07 Wilshire 0782          1    331
## 5 2.01e8 10/03/20~ 09/29/2~ 1830     14 Pacific 1454          1    420
## 6 2.40e8 12/11/20~ 11/11/2~ 1210     04 Hollenbe~ 0429          2    354
## # i 19 more variables: crm_cd_desc <chr>, mocodes <chr>, vict_age <dbl>,
## #   vict_sex <chr>, vict_descent <chr>, premis_cd <dbl>, premis_desc <chr>,
## #   weapon_used_cd <dbl>, weapon_desc <chr>, status <chr>, status_desc <chr>,
## #   crm_cd_1 <dbl>, crm_cd_2 <dbl>, crm_cd_3 <dbl>, crm_cd_4 <lgl>,
## #   location <chr>, cross_street <chr>, lat <dbl>, lon <dbl>
```

1.2 Select Columns

```
## # A tibble: 6 x 7
##   rpt_no date_occ          time_occ crime_cd case_status lat lon
##   <dbl> <chr>          <chr>    <dbl> <chr>    <dbl> <dbl>
## 1 211507896 11/07/2020 12:00:00 AM 0845      354 Invest Cont 34.2 -118.
## 2 201516622 10/18/2020 12:00:00 AM 1845      230 Invest Cont 34.2 -118.
## 3 240913563 10/30/2020 12:00:00 AM 1240      354 Invest Cont 34.2 -118.
## 4 210704711 12/24/2020 12:00:00 AM 1310      331 Invest Cont 34.0 -118.
## 5 201418201 09/29/2020 12:00:00 AM 1830      420 Invest Cont 34.0 -118.
## 6 240412063 11/11/2020 12:00:00 AM 1210      354 Invest Cont 34.1 -118.
```

1.3 Clean Time Data

1.4 Clean Coordinates & Flag Invalid Locations

```
## Records with valid coordinates:1002654
## Records with missing coordinates removed: 2240
```

1.5 Remove Duplicates

```
## Duplicate rows removed:0
## Final row count:      1002654
```

1.6 Missing Values Check

```
## Total missing values in cleaned LAPD data: 0
```

1.7 Derive Variables

1.8 Export Clean LAPD Data

Output files already exist.

2 ACS Census Data

2.1 Load Raw Data

```
## ACS data loaded from cache.

## Simple feature collection with 6 features and 5 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: -118.1524 ymin: 33.77532 xmax: -118.1227 ymax: 33.7826
## Geodetic CRS: NAD83
##      GEOID      NAME      variable
## 1 06037577000 Census Tract 5770; Los Angeles County; California median_age
## 2 06037577000 Census Tract 5770; Los Angeles County; California total_pop
## 3 06037577000 Census Tract 5770; Los Angeles County; California white
## 4 06037577000 Census Tract 5770; Los Angeles County; California black
## 5 06037577000 Census Tract 5770; Los Angeles County; California native
## 6 06037577000 Census Tract 5770; Los Angeles County; California asian
## estimate moe geometry
## 1 35.7 2.6 MULTIPOLYGON (((-118.1524 3...
## 2 7177.0 1089.0 MULTIPOLYGON (((-118.1524 3...
## 3 2477.0 482.0 MULTIPOLYGON (((-118.1524 3...
## 4 886.0 340.0 MULTIPOLYGON (((-118.1524 3...
## 5 0.0 19.0 MULTIPOLYGON (((-118.1524 3...
## 6 656.0 399.0 MULTIPOLYGON (((-118.1524 3...
```

2.2 Flag Unreliable Estimates

Note: Several variables show high rates of unreliable estimates across census tracts, particularly small subgroup counts. This is expected in small-population tracts, but should be interpreted with caution.

Variables with unreliable estimates (MOE > 50%):

Variable	Tracts with Unreliable Estimates
edu_ged	2222
single_parent_f	2143
vacant_units	2130
edu_12th	2118
labor_unemployed	2110
two_or_more	2097
housing_burden_mort	2023
edu_some_college	1965
edu_no_hs	1938
black	1868
edu_11th	1850
disability_min	1782
edu_professional	1773
edu_9th	1758
no_vehicle_renter	1746
no_vehicle_owner	1738
edu_10th	1729
edu_doctorate	1711
lang_iso_spanish	1600
edu_associates	1572
disability_mid	1558

Variable	Tracts with Unreliable Estimates
poverty	1455
lang_iso_asian	1443
single_parent_m	1319
housing_burden_rent	1314
edu_masters	1303
other_race	1278
disability_eld	1271
asian	1180
lang_iso_indoeur	904
white	820
edu_some_college_m	744
native	601
edu_hs_diploma	557
pacific_islander	450
edu_bachelors	438
hispanic	322
lang_iso_other	313
renter_units	240
median_income	115
labor_total	15
poverty_pop	15
disability_pop	14
building_units	13
lang_iso_total	13
occupied_units	13
total_housing	13
edu_total	12
internet_access	12
total_pop	11
gini	7
household_size	2
median_age	2

2.3 Reshape Data

2.4 Remove Zero Population Tracts

```
## Zero/null population tracts removed: 21
## Remaining tracts: 2477
```

2.5 Derive Variables

2.6 Check Missing Values

Variable	Missing Values
median_income	21
gini	11
household_size	8
no_vehicle_rate	8
housing_burden_rate	8
vacancy_rate	8

Variable	Missing Values
renter_rate	8
single_parent_rate	8
lang_isolation_rate	8
internet_rate	8
poverty_rate	5
unemployment_rate	3
disability_rate	3
median_age	2
edu_rate_no_hsdip	1
edu_rate_hs	1
edu_rate_some_col	1
edu_rate_college	1

2.7 Impute Missing Values

Dataset	Income Mean	Income SD	Gini Mean	Gini SD
1	75961.86	54899.34	0.4131	0.1392
2	72538.48	52582.06	0.4037	0.1445
3	79495.62	48221.30	0.4144	0.1448
4	80560.24	51331.76	0.4004	0.1374
5	70858.52	50127.60	0.4140	0.1414

Missing values after imputation: 0

2.8 Drop Raw Input Variables

```
## Raw input variables dropped: 47  
## Remaining ACS columns:      29
```

2.9 Export Clean ACS Data

```
## Output files already exist - skipping export.
```

3 Join and Aggregate Data

3.1 Convert LAPD to Spatial Points

3.2 Spatial Join

```
## Incidents matched to a tract: 992757
## Incidents with no tract match: 10007
```

3.3 Build RQ1 Incident-Level Dataset

RQ1 — *How do neighborhood-level socioeconomic characteristics predict whether a crime incident is violent or property-related?* — requires an incident-level dataset where each row is a single crime, the outcome is a binary violent/property flag, and the ACS socioeconomic characteristics of the incident's census tract serve as predictors. Drug, weapons-only, and other incidents that are neither violent nor property crimes are excluded.

```
## RQ1 incident-level dataset:
##   Total incidents (violent + property): 754509
##   Violent:                             145305
##   Property:                            609204
##   Unique tracts:                       1225
## Output files already exist - skipping export.
```


3.4 Aggregate to Tract Level for RQ2

3.5 Compute Crime Rates

3.6 Join ACS Variables

3.7 Filter to City of Los Angeles

```
## City boundary loaded from cache.
```

```
## Tracts outside City of LA removed: 1186
## Remaining tracts: 1291
```

3.8 Tract Flags

```
## Small population tracts (<500): 12
## Zero incident tracts: 59
## Extreme rate tracts (3 SD): 2
```

3.9 Check Join Results

```
## Total tracts in final dataset: 1291
## Tracts with zero incidents: 59
## Tracts with missing crime rates: 0
```

3.10 Zero-Incident Tract Details

Zero-incident tracts with small population (<500): 1 of 59

GEOID	Name	Total Population	Small Pop Flag
06037980011	Census Tract 9800.11; Los Angeles County; California	121	TRUE
06037603004	Census Tract 6030.04; Los Angeles County; California	1649	FALSE
06037113235	Census Tract 1132.35; Los Angeles County; California	1811	FALSE
06037535606	Census Tract 5356.06; Los Angeles County; California	1949	FALSE
06037300902	Census Tract 3009.02; Los Angeles County; California	2051	FALSE
06037600601	Census Tract 6006.01; Los Angeles County; California	2394	FALSE
06037482001	Census Tract 4820.01; Los Angeles County; California	2549	FALSE
06037600802	Census Tract 6008.02; Los Angeles County; California	2570	FALSE
06037602202	Census Tract 6022.02; Los Angeles County; California	2728	FALSE
06037311801	Census Tract 3118.01; Los Angeles County; California	2811	FALSE
06037311601	Census Tract 3116.01; Los Angeles County; California	2996	FALSE
06037463900	Census Tract 4639; Los Angeles County; California	3007	FALSE
06037480802	Census Tract 4808.02; Los Angeles County; California	3180	FALSE
06037302506	Census Tract 3025.06; Los Angeles County; California	3205	FALSE
06037603007	Census Tract 6030.07; Los Angeles County; California	3463	FALSE
06037800206	Census Tract 8002.06; Los Angeles County; California	3493	FALSE
06037302104	Census Tract 3021.04; Los Angeles County; California	3510	FALSE
06037601501	Census Tract 6015.01; Los Angeles County; California	3553	FALSE
06037600911	Census Tract 6009.11; Los Angeles County; California	3595	FALSE
06037572301	Census Tract 5723.01; Los Angeles County; California	3648	FALSE
06037302301	Census Tract 3023.01; Los Angeles County; California	3654	FALSE
06037800104	Census Tract 8001.04; Los Angeles County; California	3728	FALSE
06037544002	Census Tract 5440.02; Los Angeles County; California	3758	FALSE
06037535001	Census Tract 5350.01; Los Angeles County; California	3804	FALSE
06037800103	Census Tract 8001.03; Los Angeles County; California	3818	FALSE
06037603008	Census Tract 6030.08; Los Angeles County; California	3820	FALSE
06037480703	Census Tract 4807.03; Los Angeles County; California	3883	FALSE
06037620102	Census Tract 6201.02; Los Angeles County; California	4014	FALSE
06037480704	Census Tract 4807.04; Los Angeles County; California	4087	FALSE
06037700502	Census Tract 7005.02; Los Angeles County; California	4087	FALSE
06037700801	Census Tract 7008.01; Los Angeles County; California	4176	FALSE
06037620001	Census Tract 6200.01; Los Angeles County; California	4242	FALSE
06037480602	Census Tract 4806.02; Los Angeles County; California	4306	FALSE
06037532302	Census Tract 5323.02; Los Angeles County; California	4437	FALSE
06037540300	Census Tract 5403; Los Angeles County; California	4479	FALSE
06037601402	Census Tract 6014.02; Los Angeles County; California	4551	FALSE
06037600702	Census Tract 6007.02; Los Angeles County; California	4704	FALSE
06037700600	Census Tract 7006; Los Angeles County; California	4721	FALSE
06037670604	Census Tract 6706.04; Los Angeles County; California	4764	FALSE
06037600201	Census Tract 6002.01; Los Angeles County; California	4830	FALSE
06037650903	Census Tract 6509.03; Los Angeles County; California	4888	FALSE
06037572600	Census Tract 5726; Los Angeles County; California	4929	FALSE
06037620101	Census Tract 6201.01; Los Angeles County; California	5050	FALSE
06037302302	Census Tract 3023.02; Los Angeles County; California	5134	FALSE
06037540800	Census Tract 5408; Los Angeles County; California	5331	FALSE
06037301000	Census Tract 3010; Los Angeles County; California	5390	FALSE
06037578000	Census Tract 5780; Los Angeles County; California	5403	FALSE
06037300701	Census Tract 3007.01; Los Angeles County; California	5444	FALSE
06037702801	Census Tract 7028.01; Los Angeles County; California	5479	FALSE

GEOID	Name	Total Population	Small Pop Flag
06037531301	Census Tract 5313.01; Los Angeles County; California	5563	FALSE
06037481901	Census Tract 4819.01; Los Angeles County; California	5581	FALSE
06037800101	Census Tract 8001.01; Los Angeles County; California	5646	FALSE
06037920312	Census Tract 9203.12; Los Angeles County; California	5684	FALSE
06037311700	Census Tract 3117; Los Angeles County; California	5763	FALSE
06037310100	Census Tract 3101; Los Angeles County; California	5968	FALSE
06037601302	Census Tract 6013.02; Los Angeles County; California	6377	FALSE
06037543306	Census Tract 5433.06; Los Angeles County; California	6708	FALSE
06037920326	Census Tract 9203.26; Los Angeles County; California	6971	FALSE
06037535101	Census Tract 5351.01; Los Angeles County; California	7219	FALSE

3.11 Remove Zero-Incident and Small Population Tracts

Tracts removed: 70

Remaining tracts: 1221

4 Variable Selection

4.1 Crime Variables

```
## Variables dropped:      28
## Remaining crime variables: 8
```

4.2 ACS Variables

4.2.1 Check Near Zero Variance

Variable	Variance	SD	Mean	Proportion Near Zero
pct_native	0.000000e+00	0.0046	0.0015	0.9550
pct_pacific	0.000000e+00	0.0053	0.0013	0.9607
pct_other	2.000000e-04	0.0129	0.0069	0.7772
pct_two_or_more	1.100000e-03	0.0329	0.0389	0.2072
unemployment_rate	1.800000e-03	0.0427	0.0817	0.0098
disability_rate	2.500000e-03	0.0503	0.1161	0.0016
single_parent_rate	2.600000e-03	0.0514	0.0595	0.1450
vacancy_rate	2.800000e-03	0.0525	0.0702	0.0688
internet_rate	4.000000e-03	0.0634	0.9240	0.0000
gini	4.100000e-03	0.0642	0.4557	0.0000
edu_rate_some_col	4.900000e-03	0.0697	0.2249	0.0000
edu_rate_hs	7.100000e-03	0.0842	0.1844	0.0033
housing_burden_rate	7.900000e-03	0.0888	0.2544	0.0016
edu_rate_no_hsdip	8.600000e-03	0.0928	0.1255	0.0688
no_vehicle_rate	1.120000e-02	0.1058	0.1174	0.0500
poverty_rate	1.120000e-02	0.1059	0.1629	0.0016
lang_isolation_rate	1.320000e-02	0.1147	0.1421	0.0541
pct_asian	1.330000e-02	0.1153	0.1208	0.1155
pct_black	1.390000e-02	0.1179	0.0823	0.1654
edu_rate_college	5.260000e-02	0.2294	0.3837	0.0000
pct_white	6.090000e-02	0.2469	0.2908	0.0786
renter_rate	6.850000e-02	0.2616	0.6210	0.0000
pct_hispanic	8.080000e-02	0.2843	0.4574	0.0008
household_size	5.759000e-01	0.7589	2.7467	0.0000
median_age	3.835890e+01	6.1935	38.3192	0.0000
building_units	3.773436e+05	614.2830	1453.9189	0.0008
total_pop	1.431227e+06	1196.3392	3599.9664	0.0000
median_income	1.769116e+09	42060.8633	90291.0925	0.0000

4.2.2 Check Highly Correlated Variables

Highly correlated pairs ($|r| > 0.70$):

Variable 1	Variable 2	Correlation
edu_rate_college	pct_hispanic	-0.886
edu_rate_college	edu_rate_no_hsdip	-0.867
edu_rate_no_hsdip	pct_hispanic	0.864
edu_rate_college	edu_rate_hs	-0.836
pct_hispanic	pct_white	-0.833
edu_rate_college	pct_white	0.832
edu_rate_no_hsdip	pct_white	-0.765
household_size	pct_hispanic	0.754
edu_rate_college	household_size	-0.716

4.2.3 Select ACS Variables

```
## ACS variables dropped:      8
## Remaining ACS predictors: 20
```

4.3 Export Dataset

```
## Output files already exist - skipping export.
```


5 Final Dataset Overview

```
## incident_data (incident-level, RQ1):  
##   Rows:      754509  
##   Columns: 30  
##  
## tract_data (tract-level, RQ2):  
##   Rows:      1221  
##   Columns: 34
```

5.1 Tract Flags

Variable	Description
flag_small_pop	Tract population below 500 (crime rates may be unstable)
flag_zero_crime	Tract had zero matched incidents 2020–2024
flag_extreme_rate	Total crime rate exceeds 3 standard deviations above the mean

Note: 58 of 59 zero-crime tracts have populations above 500, suggesting these zeros reflect a spatial join artifact at the city boundary rather than genuine absence of crime. These tracts were excluded from the final dataset. Tracts with fewer than 500 residents were also excluded due to unstable crime rates.

5.2 Tract-Level Crime Variables (`tract_data`)

Variable	Type	Description
Spatial Analysis Outcomes		
total_incidents	Count	Total incidents 2020–2024
n_violent	Count	Total violent crime incidents
Spatial Analysis Rates		
rate_total	Rate (per 1,000)	Total incident rate
rate_violent	Rate (per 1,000)	Violent incident rate
rate_property	Rate (per 1,000)	Property crime rate
rate_drug	Rate (per 1,000)	Drug/narcotic incident rate
Descriptive		
n_arrested	Count	Total arrests
n_cleared_other	Count	Total cases closed for other reasons

5.3 RQ1 Incident-Level Variables (`incident_data`)

Variable	Description
Incident Identifiers	
<code>rpt_no</code>	LAPD report number
<code>date_occ</code>	Date of occurrence
Crime Classification	
<code>crime_cd</code>	LAPD crime code
<code>crime_category</code>	Detailed crime category
<code>crime_type</code>	Violent / Property
Outcome Variables	
<code>is_violent</code>	TRUE if violent crime
<code>is_property</code>	TRUE if property crime
<code>outcome</code>	Binary outcome: 1 = violent, 0 = property
<code>outcome_multi</code>	Multiclass outcome: specific crime category as factor
Geography	
<code>GEOID</code>	Census tract GEOID (links to ACS predictors)
Economic Vulnerability	
<code>median_income</code>	Median household income
<code>gini</code>	Gini coefficient
<code>poverty_rate</code>	Proportion population below poverty line
<code>unemployment_rate</code>	Proportion labor force unemployed
<code>no_vehicle_rate</code>	Proportion households with no vehicle
<code>housing_burden_rate</code>	Proportion 50%+ of income on housing
Neighborhood Stability	
<code>vacancy_rate</code>	Proportion housing units vacant
<code>renter_rate</code>	Proportion housing units renter-occupied
Social Disorganization	
<code>single_parent_rate</code>	Proportion households with single parent
<code>lang_isolation_rate</code>	Proportion with limited English proficiency
<code>internet_rate</code>	Proportion households with internet access
Demographics	
<code>total_pop</code>	Total tract population
<code>median_age</code>	Median age of residents
<code>pct_white</code>	Proportion non-Hispanic white
<code>pct_black</code>	Proportion non-Hispanic Black
<code>pct_asian</code>	Proportion non-Hispanic Asian
<code>pct_hispanic</code>	Proportion Hispanic or Latino
Education & Disability	
<code>edu_rate_no_hsdip</code>	Proportion with less than high school diploma
<code>edu_rate_college</code>	Proportion with bachelor's degree or higher
<code>disability_rate</code>	Proportion with a disability

5.4 Tract-Level ACS Variables (`tract_data`)

Variable	Description
Economic Vulnerability	
median_income	Median household income
gini	Gini coefficient (0 = equal, 1 = unequal)
poverty_rate	Proportion population below poverty line
unemployment_rate	Proportion labor force unemployed
no_vehicle_rate	Proportion households with no vehicle
housing_burden_rate	Proportion 50%+ of income on housing
Neighborhood Stability	
vacancy_rate	Proportion housing units vacant
renter_rate	Proportion housing units renter-occupied
Social Disorganization	
single_parent_rate	Proportion households with single parent
lang_isolation_rate	Proportion with limited English proficiency
internet_rate	Proportion households with internet access
Demographics	
total_pop	Total tract population
median_age	Median age of residents
pct_white	Proportion non-Hispanic white
pct_black	Proportion non-Hispanic Black
pct_asian	Proportion non-Hispanic Asian
pct_hispanic	Proportion Hispanic or Latino
Education	
edu_rate_no_hsdip	Proportion with less than high school diploma
edu_rate_college	Proportion with bachelor's degree or higher
Disability	
disability_rate	Proportion with a disability